

SDET Skillset Matrix

Overview

The **SDET (Software Development Engineer in Test) Skills Matrix** is designed to establish a **clear framework** for assessing and improving the technical competencies of the **Automation & Performance Testing Team**. It serves as a tool for **career development, resource management, project planning, and identifying skill gaps**.

This page provides:

- A **structured skills matrix** with categorized competencies required for an SDET.
- Guidance for **self-assessment & tracking progress**.
- Insights into **team strengths & areas for improvement**.
- A standardized **career growth & training roadmap**.

Why a Skills Matrix?

1 Career Development

Empower Growth: Team members can identify skill improvement areas and focus on acquiring relevant competencies.
Structured Progression: Helps define clear **learning paths** aligned with project needs.

2 Identifying Skill Gaps

Strength vs. Gaps: Compare individual skills with **required expertise** to pinpoint areas for **training or hiring**.
Targeted Learning: Design team-wide **upskilling initiatives** to bridge competency gaps.

3 Resource Management

Optimize Team Strengths: Ensure project assignments align with **team members' expertise**.
Upskill Before Hiring: Identify whether **training** or **new hires** are needed.

4 Project Planning & Allocation

Role-Based Allocation: Assign tasks to **best-fit team members** based on their technical proficiencies.
Balanced Workload Distribution: Improve efficiency & team **collaboration by leveraging diverse skill sets**.

SDET Required Skill Set

Programming Languages

Skill	Requirement
Java	Must Have
C#	Nice to Have
Python	Optional
JavaScript	Nice to Have
PowerShell	Optional

Web Technologies

Skill	Requirement
HTML, CSS	Nice to Have

Test Automation Frameworks

Skill	Requirement
Selenium	Must Have
Selenium Grid	Nice to Have
Cucumber (Gherkin)	Must Have
BrowserStack	Nice to Have
TestNG	Must Have
Maven	Nice to Have
JUnit	Nice to Have

API Testing

Skill	Requirement
Rest Assured	Must Have
Postman	Must Have
SoapUI	Optional

Performance Testing

Skill	Requirement
Performance Testing Concepts	Nice to Have
JMeter	Nice to Have
Blazemeter	Nice to Have
Taurus	Nice to Have

Database

Skill	Requirement
Oracle	Must Have
PL-SQL	Nice to Have

CI/CD & DevOps

Skill	Requirement
Jenkins	Nice to Have
CI/CD Pipelines	Nice to Have
Docker/Kubernetes	Nice to Have
Azure DevOps	Nice to Have

Agile & Collaboration

Skill	Requirement
Agile Methodologies	Must Have
TDD, BDD	Must Have

Mobile Testing

Skill	Requirement
Appium	Nice to Have

Bug & Defect Tracking

Skill	Requirement
JIRA or Similar	Must Have
QTest	Nice to Have

Version Control

Skill	Requirement
GitLab or Git	Must Have

Cloud & Infrastructure

Skill	Requirement
AWS/Azure	Optional
Shell Scripting	Optional

Monitoring & AI Tools

Skill	Requirement
Splunk	Optional
GitHub Copilot	Optional

How to Use the Skills Matrix

Step 1: Self-Assessment

Download the SDET Skills Matrix File: [SDET Skills Matrix.xlsx](#)

Update your details – Mark your current proficiency level for each skill.

Rate yourself from 1-5 (1 = Beginner, 5 = Expert).

Step 2: Submit to Your Reporting Manager

Email your completed skills matrix to your **reporting manager** for review.

Step 3: Continuous Learning & Growth

Identify **skill gaps** and **focus on upskilling** through:

- Internal **training sessions & workshops**
- Online **courses & certifications**
- Hands-on **project experience**

Mentorship Opportunities: Collaborate with **senior SDETs** to gain insights on complex automation frameworks.

Next Steps

All team members should **update their skills matrix** quarterly.

Use the skills matrix for **performance reviews** and career growth discussions.

Leverage training programs to enhance missing competencies.

For questions or assistance, contact: [Team Leads/Managers]

Stay ahead in automation testing – Continuous Learning = Growth!

Please use the file [SDET Skills Matrix.xlsx](#), update with your details (self-assessment) and email to your reporting manager.